

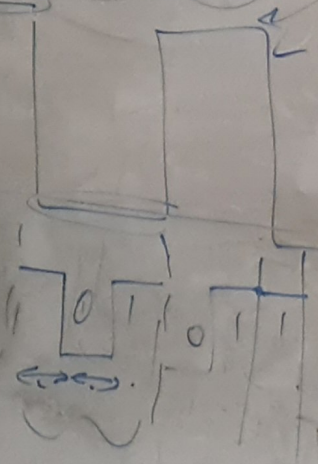
Lec. 2 AM [Amplitude Mod.] Conventional Mod.

→ Analog. / Digital

5V → 101
High 5V → 011

101

0V

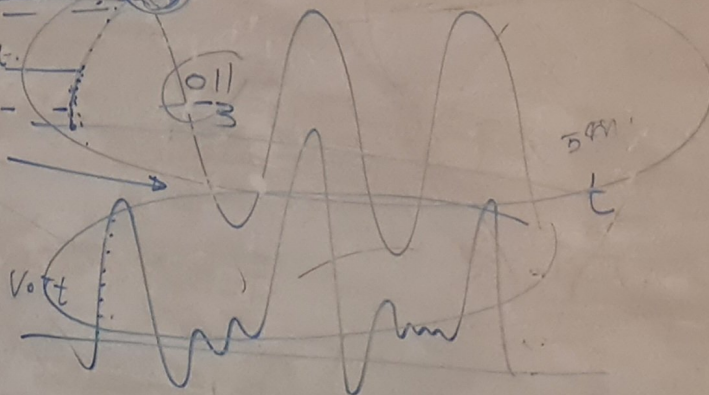


Analog

volt 101 5V

volt

011
3



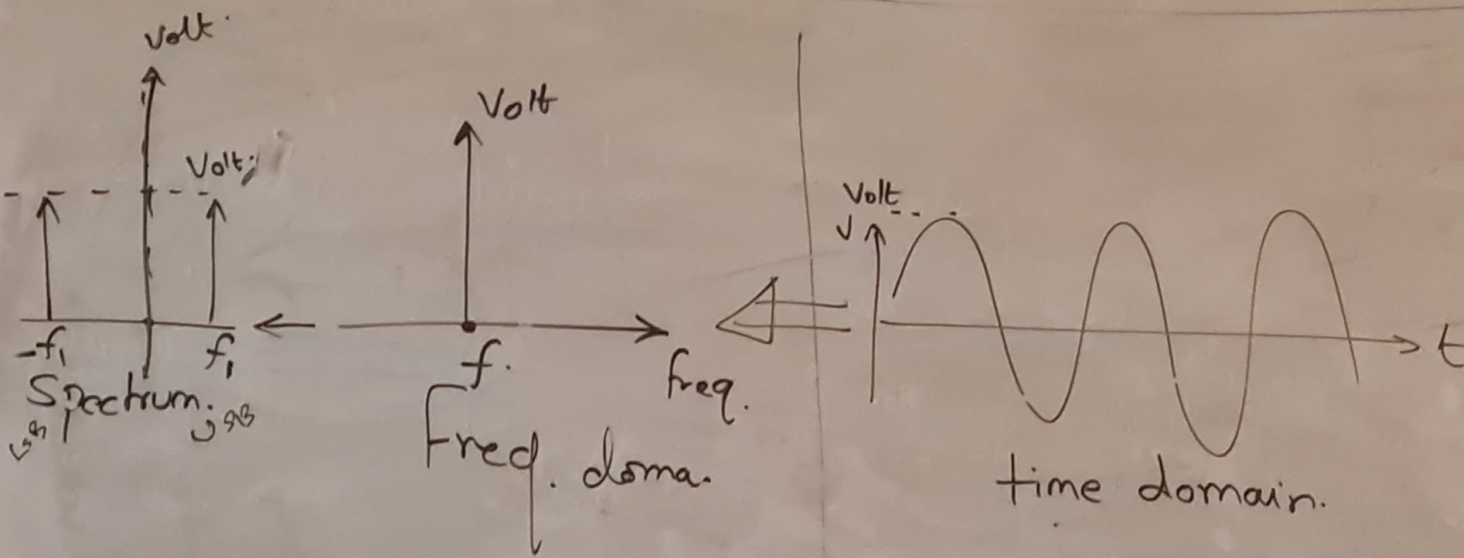
Analog Mod.

AM

FM/PM

Lec. 2 AM [Amplitude Mod.]

Conventional Mod.

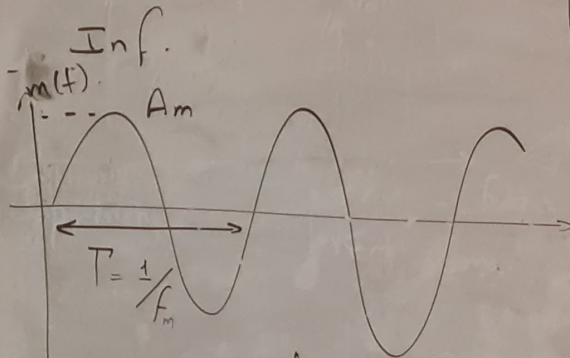


AM

Lec. 2 AM [Amplitude Mod.]

$V_c(t)$ Conventional Mod.

Amplitude Mod.



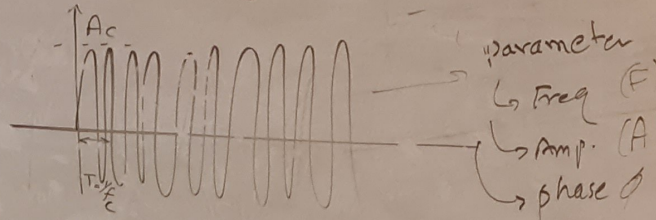
Information Signal

$$m(t) = A_m \cos(\omega_m t)$$

$$m(t) = A_m \cos(2\pi f_m t)$$

$f_m \rightarrow$ is very small.

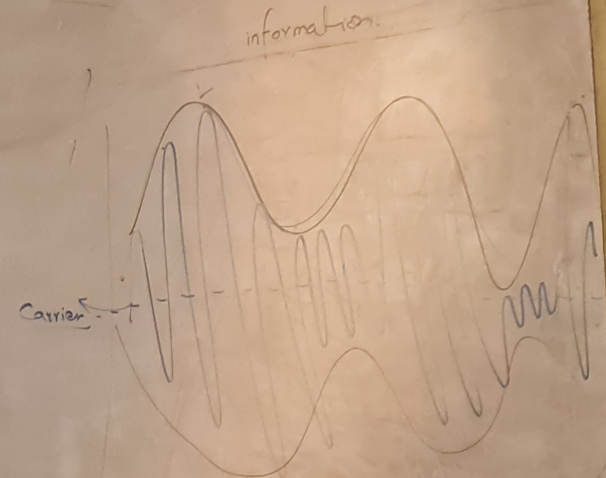
Carrier Signal. (crystal).
 $c(t)$ oscillator



$$c(t) = A_c \cos(\omega_c t)$$

$$= A_c \cos(2\pi f_c t)$$

Where f_c is High. (kHz MHz GHz)

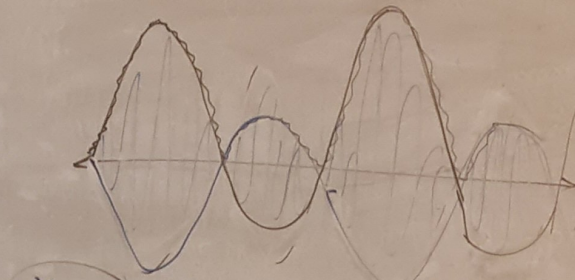


Lec. 2 AM [Amplitude Mod.]

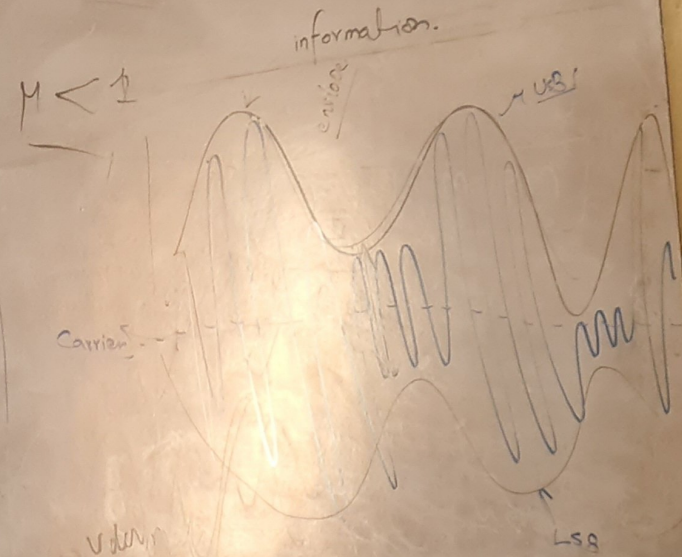
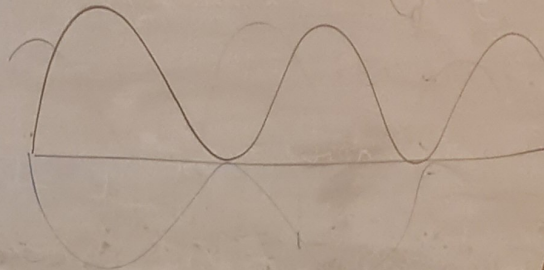
$V_c(t)$ Conventional Mod.

$\mu \rightarrow$ Modulation Index

$\mu > 1$	$\rightarrow$ over mod.
$\mu = 1$	$\rightarrow$ critical Mod.
$\mu < 1$	under mod.



critical Mod.



Lec.2 AM [Amplitude Mod.]

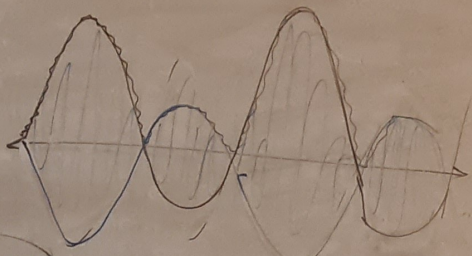
$V_c(t)$ Conventional Mod.

$M \rightarrow$ Modulation Index

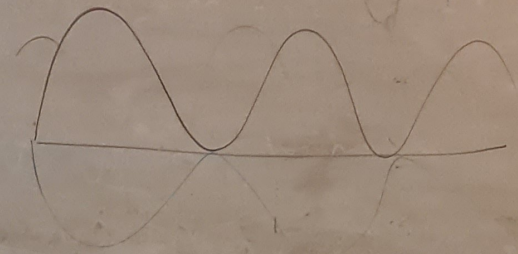
- $\rightarrow$ over mod.
- $\rightarrow$ critical Mod.
- $\frac{1}{1}$ under mod.

$\frac{A_m}{A_c}$

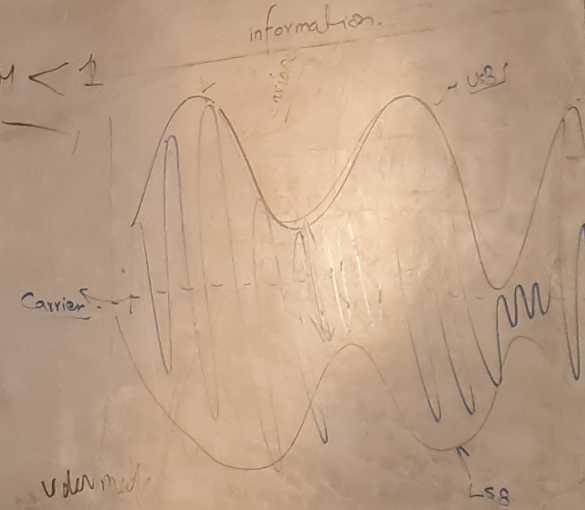
critical Mod.



$M > 1$



$M < 1$



under mod.

LSB

Lec. 2 AM [Amplitude Mod.]

Conventional Mod.

$$S(t) = A_c [1 + \mu \cos 2\pi f_m t] \cos 2\pi f_c t$$

$$= A_c \cos 2\pi f_c t + A_c \mu \cos 2\pi f_c t \cos 2\pi f_m t$$

$$S_{AM}(t) = \underbrace{A_c \cos 2\pi f_c t}_{\text{Carrier}} + \underbrace{\frac{A_c \mu}{2} \cos(2\pi(f_c + f_m)t)}_{\text{USB}} + \underbrace{\frac{A_c \mu}{2} \cos(2\pi(f_c - f_m)t)}_{\text{LSB}}$$

$$m(t) = A_m \cos \omega_m t$$

$$c(t) = A_c \cos \omega_c t$$

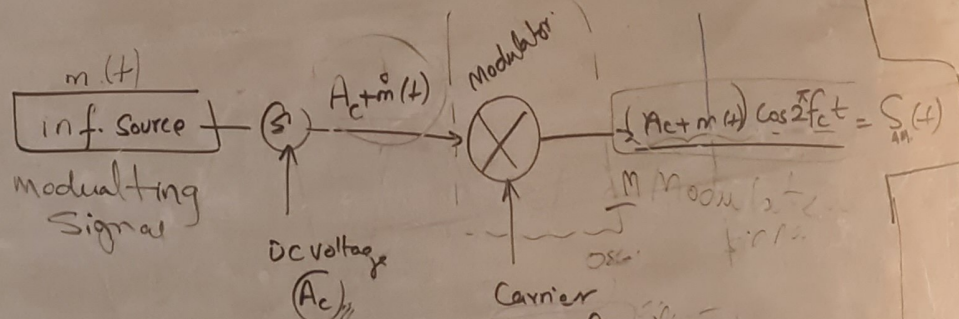
$$S_{AM}(t) = [A_c + A_m \cos(2\pi f_m t)] \cos 2\pi f_c t$$

$$= A_c \left[1 + \left(\frac{A_m}{A_c} \right) \cos 2\pi f_m t \right] \cos 2\pi f_c t$$

$$\frac{A_m}{A_c} < 1$$

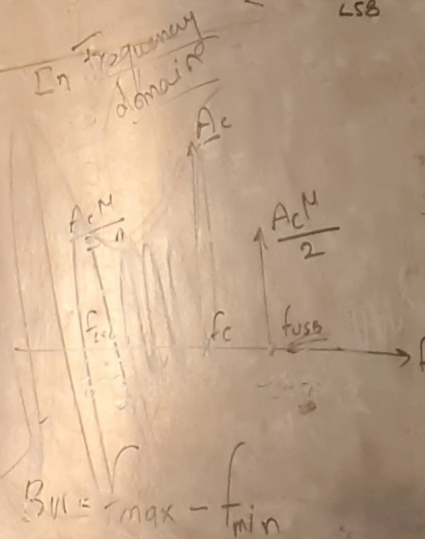
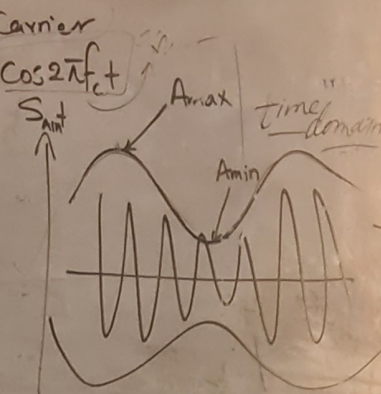
Modulation Index

$$S_{AM}(t) = A_c [1 + \mu \cos 2\pi f_m t] \cos 2\pi f_c t$$



Other equation

$$\mu = \frac{A_{max} - A_{min}}{A_{max} + A_{min}}$$



$$B_{AM} = f_{max} - f_{min}$$

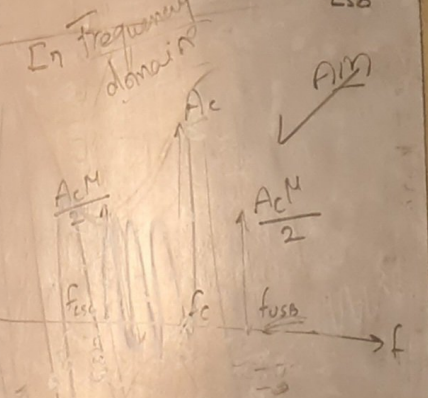
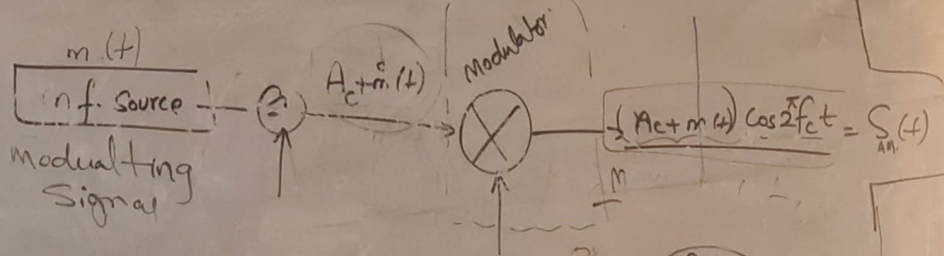
Lec.2 AM [Amplitude Mod.]

$V_c(t)$ Conventional Mod.

$$S(t) = A_c [1 + \mu \cos 2\pi f_m t] \cos 2\pi f_c t$$

$$= A_c \cos 2\pi f_c t + A_c \mu \cos 2\pi f_c t \cos 2\pi f_m t$$

$$S_{AM}(t) = \underbrace{A_c \cos 2\pi f_c t}_{\text{Carrier}} + \underbrace{\frac{A_c \mu}{2} \cos(2\pi(f_c + f_m)t)}_{\text{USB}} + \underbrace{\frac{A_c \mu}{2} \cos(2\pi(f_c - f_m)t)}_{\text{LSB}}$$



$$P_t = P_c + P_{USB} + P_{LSB}$$

$$= \frac{A_c^2}{2R} + \frac{A_c^2 \mu^2}{8R} + \frac{A_c^2 \mu^2}{8R}$$

$$P_{SB}$$

$$P_c = \frac{A_c^2}{2R}$$

$$P_{USB} = \frac{A_c^2 \mu^2}{8R}$$

$$P_{LSB} = \frac{A_c^2 \mu^2}{8R}$$

$$P_{SB} = \frac{P_c \mu^2}{2}$$

$$P_c = 2P_{SB}$$

$$P_t = P_c \left(1 + \frac{\mu^2}{2}\right)$$

Efficiency

$$\eta = \frac{\mu^2}{2 + \mu^2} \times 100$$

$$\eta = \frac{P_{SB}}{P_t}$$

$$\mu = 1$$

$$\eta = 33.33\%$$

$$BW = f_{max} - f_{min}$$

$$= (f_c + f_m) - (f_c - f_m) = 2f_m$$

Lec. 2 AM [Amplitude Mod.]

$V_c(t)$ Conventional Mod.

$$S(t) = A_c [1 + \mu \cos 2\pi f_m t] \cos 2\pi f_c t$$

$$= A_c \cos 2\pi f_c t + A_c \mu \cos 2\pi f_c t \cos 2\pi f_m t$$

A modulating signal $(10 \cos(2\pi \times 10^3 t))$ is amplitude Modulated.

with carrier $(50 \times \cos(2\pi \times 10^5 t))$, Find the: 1] Mod. Index μ , 2] P_c 3] P_t

$m(t) = 10 \cos(2\pi \times 10^3 t)$

$C(t) = 50 \times \cos(2\pi \times 10^5 t)$

$A_m = 10V, f_m = 10^3 Hz$

$A_c = 50V, f_c = 10^5 Hz$

$\mu = \frac{A_m}{A_c} = \frac{10}{50} = 0.25V < 1 m(t)$

$P_c = \frac{A_c^2}{2R} = \frac{50^2}{2 \times 1} = 1250W$

$P_t = P_c (1 + \frac{\mu^2}{2}) = 1250 (1 + \frac{0.25^2}{2}) = 1281.25W$

Watts carrier $\cos 2\pi f_c t$

$S(t)$